

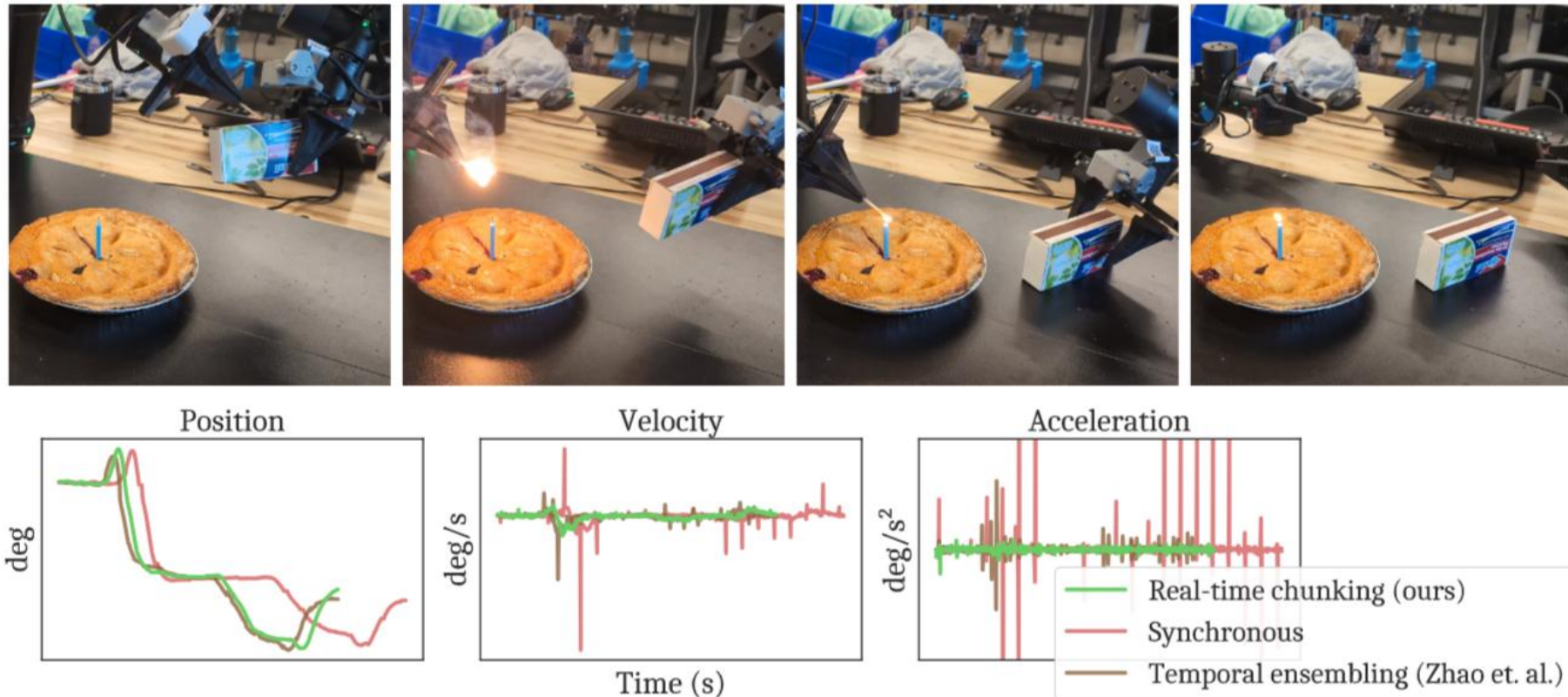


Real-Time Execution of Action Chunking Flow Policies (NeurIPS 2026)

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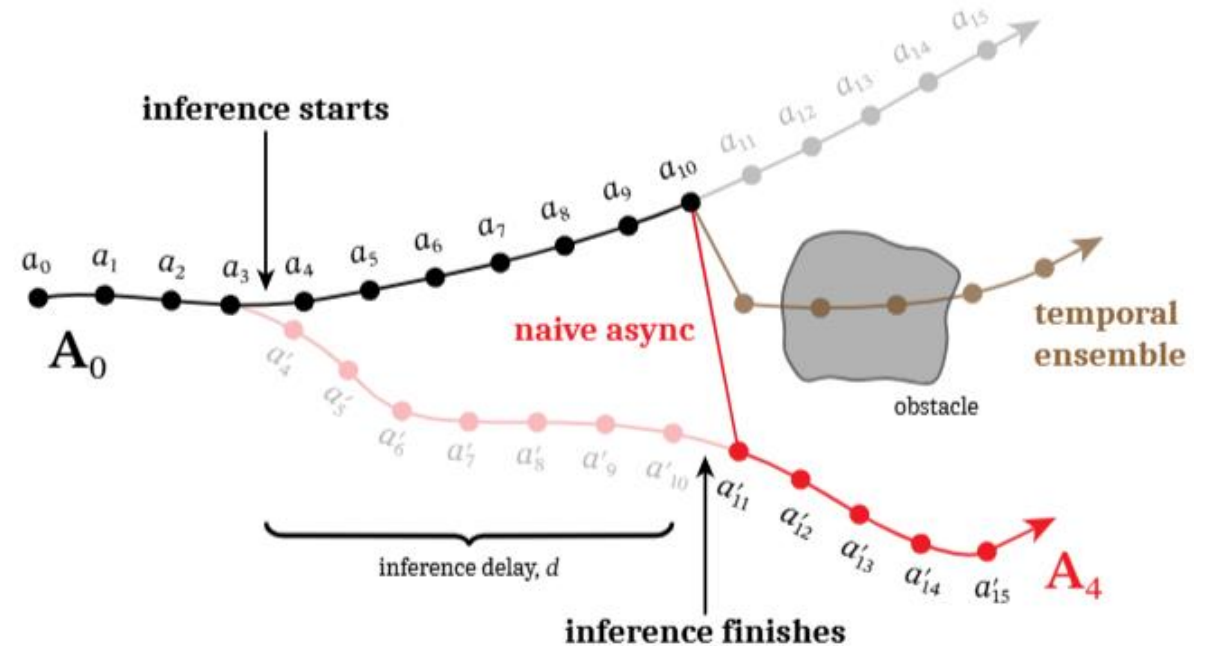
Motivation

- VLA의 높은 지연 시간으로 인해 action이 멈추거나 끊어짐
- Action chunk 기반 모델에서 다음 chunk로 넘어갈 때 동작이 어색해짐



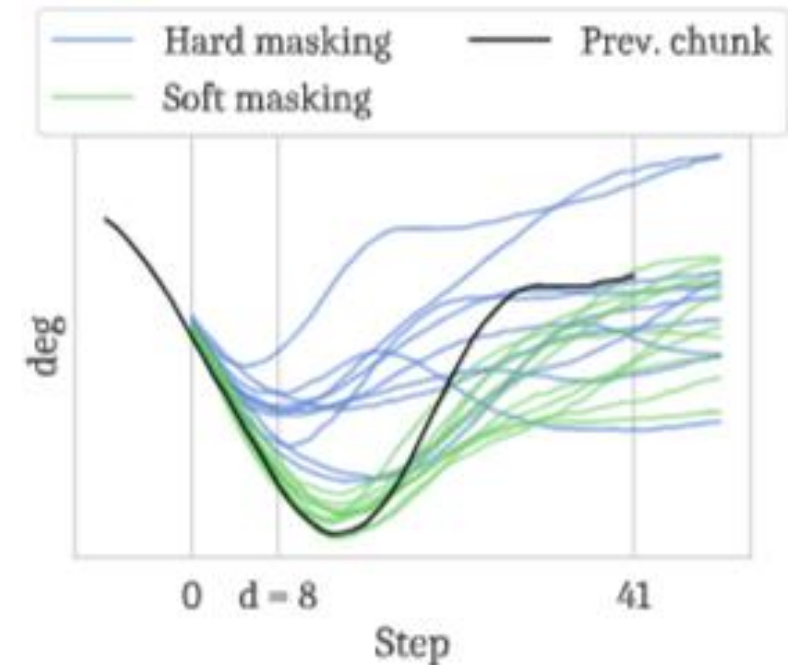
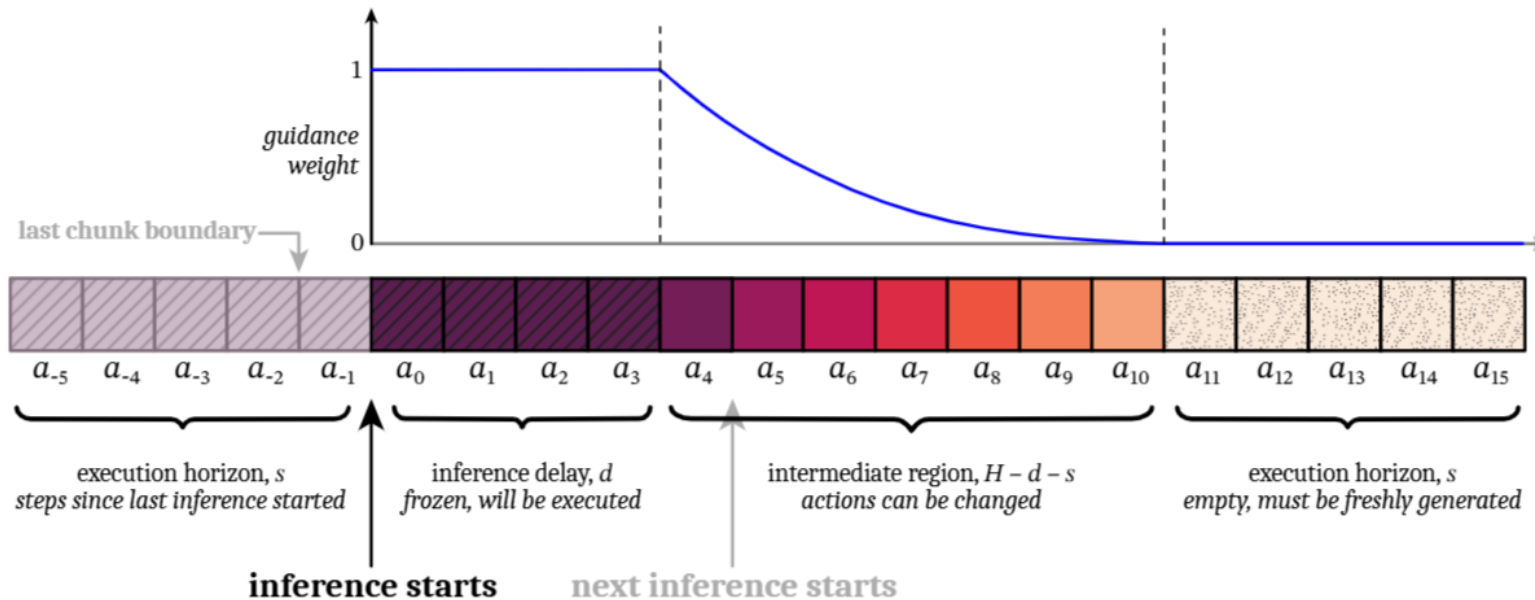
Motivation

- Naïve Synchronous: next chunk가 준비될 때까지 로봇이 멈춰야 함
- Naïve Asynchronous: new chunk가 시작되는 시점과 이전 동작이 불연속적임
- Temporal Ensembling: Multimodality를 가진 데이터에서 잘못된 평균 action을 생성할 수 있음
 - Temporal Ensembling: Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware(ICML 2023 workshop)



Method

■ 실시간 action chunking



Method

2개의 스레드를 사용

■ GETACTION

- 로봇의 관측 정보 업데이트 및 다음 행동 신호를 보냄
- 현재 action을 실행

■ INFERENCELOOP

- s_{min} 번만큼 기존 action을 실행할 때까지 기다렸다가 남은 미래 궤적을 추론
- 최근의 추론 delay를 확인하여 다음 추론 시간 예측
- 추론하는 동안 GETACTION은 기존 chunk 수행
- 추론이 끝나면 새로운 action chunk로 swap

■ GUIDEDINFERENCE

- Denoising

Algorithm 1 Real-Time Chunking

Require: flow policy π with prediction horizon H , minimum execution horizon s_{min} , mutex $\mathcal{M}$, condition variable $\mathcal{C}$ associated with $\mathcal{M}$, initial chunk $\mathbf{A}_{init}$, initial delay estimate d_{init} , delay buffer size b , number of denoising steps n , maximum guidance weight β

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1: procedure INITIALIZE_SHARED_STATE ▷ Initialize mutex-protected shared variables
2:    $t = 0$ ;  $\mathbf{A}_{cur} = \mathbf{A}_{init}$ ,  $\mathbf{o}_{cur} = \text{null}$ 

3: function GETACTION( $\mathbf{o}_{next}$ ) ▷ Called at an interval of  $\Delta t$  by controller
4:   with  $\mathcal{M}$  acquired do
5:      $t = t + 1$ 
6:      $\mathbf{o}_{cur} = \mathbf{o}_{next}$ 
7:     notify  $\mathcal{C}$ 
8:   return  $\mathbf{A}_{cur}[t - 1]$ 
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9: procedure INFERENCELOOP ▷ Run inference in a looping background thread
10:  acquire  $\mathcal{M}$ 
11:   $\mathcal{Q} = \text{new Queue}([d_{init}], \text{maxlen}=b)$  ▷ Holds a limited buffer of past inference delays
12:  loop
13:    wait on  $\mathcal{C}$  until  $t \geq s_{min}$ 
14:     $s = t$  ▷  $s$  is the number of actions executed since last inference started
15:     $\mathbf{A}_{prev} = \mathbf{A}_{cur}[s, s + 1, \dots, H - 1]$  ▷ Remove the  $s$  actions that have already been executed
16:     $\mathbf{o} = \mathbf{o}_{cur}$ 
17:     $d = \max(\mathcal{Q})$  ▷ Estimate the next inference delay conservatively
18:    with  $\mathcal{M}$  released do
19:       $\mathbf{A}_{new} = \text{GUIDEDINFERENCE}(\pi, \mathbf{o}, \mathbf{A}_{prev}, d, s)$ 
20:       $\mathbf{A}_{cur} = \mathbf{A}_{new}$  ▷ Swap to the new chunk as soon as it is available
21:       $t = t - s$  ▷ Reset  $t$  so that it indexes into  $\mathbf{A}_{new}$ 
22:      enqueue  $t$  onto  $\mathcal{Q}$  ▷ Record the observed delay
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23: function GUIDEDINFERENCE( $\pi, \mathbf{o}, \mathbf{A}_{prev}, d, s$ )
24:  compute  $\mathbf{W}$  using Eq. 5; right-pad  $\mathbf{A}_{prev}$  to length  $H$ ; initialize  $\mathbf{A}^0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
25:  for  $\tau = 0$  to 1 with step size  $1/n$  do
26:     $f_{\mathbf{A}^1} = \mathbf{A}' \mapsto \mathbf{A}' + (1 - \tau)\mathbf{v}_{\pi}(\mathbf{A}', \mathbf{o}, \tau)$  ▷ Define denoising function (Eq. 3)
27:     $\mathbf{e} = (\mathbf{A}_{prev} - f_{\mathbf{A}^1}(\mathbf{A}^\tau))^T \text{diag}(\mathbf{W})$  ▷ Weighted error term from Eq. 2
28:     $\mathbf{g} = \mathbf{e} \cdot \frac{\partial f_{\mathbf{A}^1}}{\partial \mathbf{A}^\tau} \Big|_{\mathbf{A}' = \mathbf{A}^\tau}$  ▷ Compute vector-Jacobian product from Eq. 2 via autodiff
29:     $\mathbf{A}^{\tau + \frac{1}{n}} = \mathbf{A}^\tau + \frac{1}{n} \left( \mathbf{v}_{\pi}(\mathbf{A}^\tau, \mathbf{o}, \tau) + \min \left( \beta, \frac{1 - \tau}{\tau \cdot r_{\tau}^2} \right) \mathbf{g} \right)$  ▷ Integration step (Eq. 1)
  return  $\mathbf{A}^1$ 
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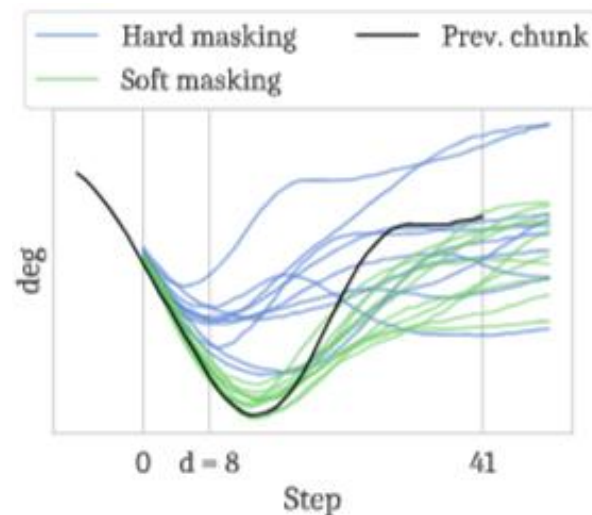
Method

■ Inference-Time Inpainting with Flow Matching

$$\mathbf{v}_{\text{HIGDM}}(\mathbf{A}_t^\tau, \mathbf{o}_t, \tau) = \mathbf{v}(\mathbf{A}_t^\tau, \mathbf{o}_t, \tau) + \min\left(\beta, \frac{1-\tau}{\tau \cdot r_\tau^2}\right) (\mathbf{Y} - \widehat{\mathbf{A}}_t^1)^\top \text{diag}(\mathbf{W}) \frac{\partial \widehat{\mathbf{A}}_t^1}{\partial \mathbf{A}_t^\tau}$$

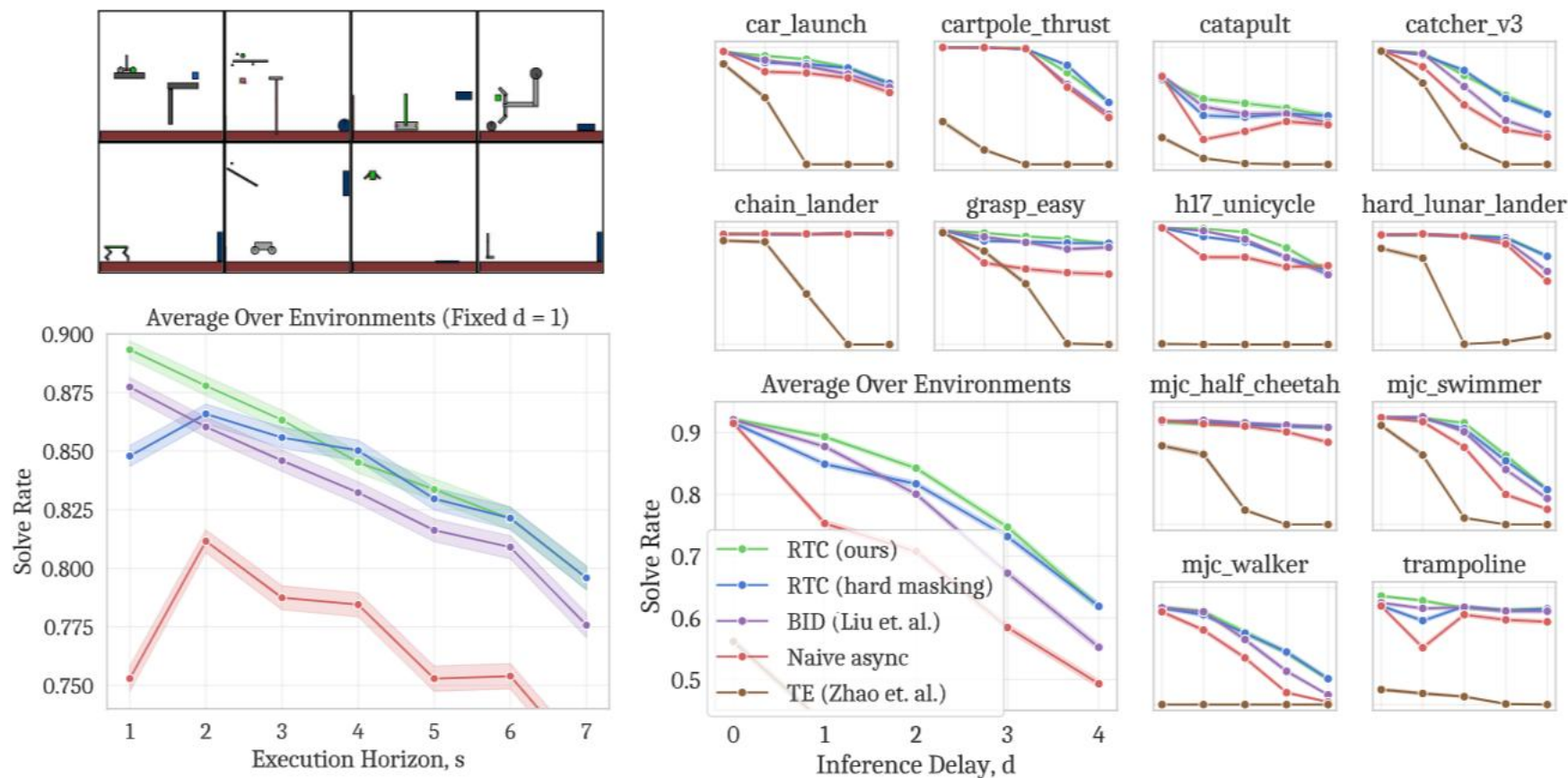
■ Soft Masking for Improved Cross-Chunk Continuity

$$\mathbf{W}_i = \begin{cases} 1 & \text{if } i < d \\ c_i \frac{e^{c_i}-1}{e-1} & \text{if } d \leq i < H-s \\ 0 & \text{if } i \geq H-s \end{cases} \quad \text{where } c_i = \frac{H-s-i}{H-s-d+1}, \quad i \in \{0, \dots, H-1\}.$$



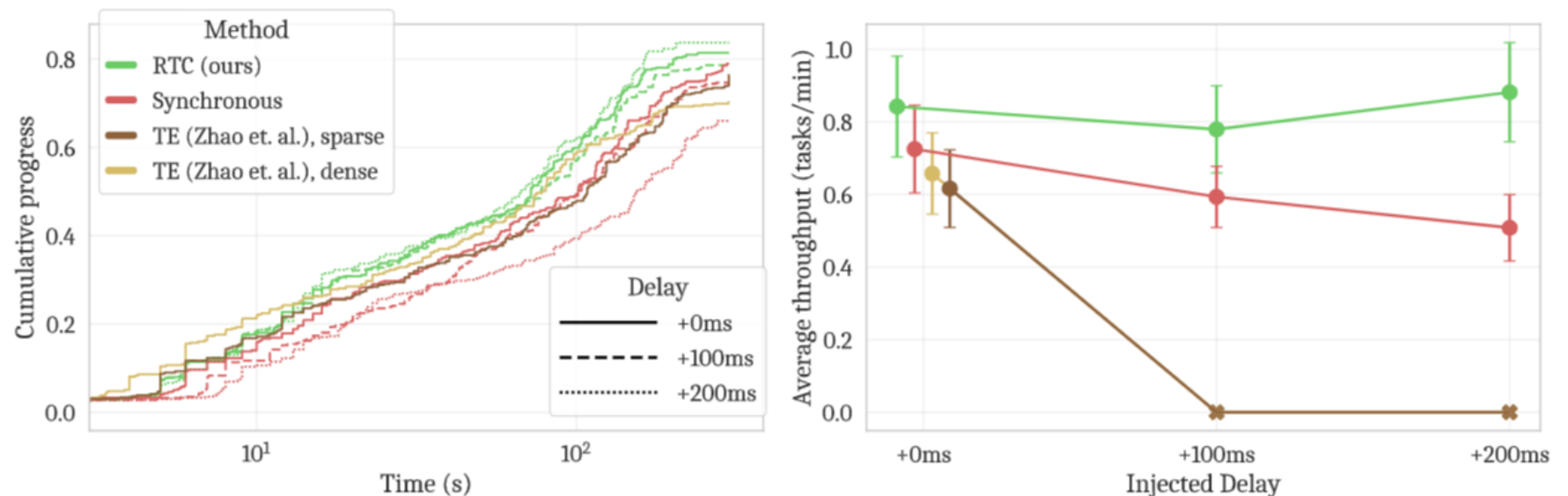
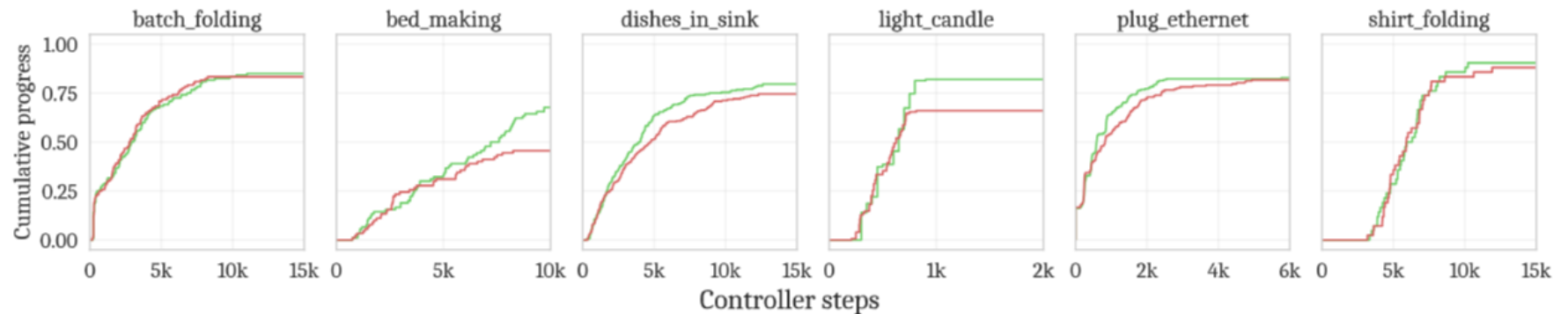
Evaluation

- Execution horizon vs. solve rate with a fixed inference delay of 1
- Inference delay vs. solve rate with a fixed execution horizon of $s = \max(d, 1)$



Evaluation

- Controller steps (equivalent to elapsed time with inference pauses removed multiplied by 50Hz) vs. cumulative progress for each task, aggregated across all delays





Thank You!